

# Architecting Modern Data Systems

A structural reference guide to NoSQL paradigms, multi-modal convergence, and scalable database selection.

Based on COM 769 Scalable Advanced Software Systems | Joe Rafferty



## Structured Data

Strict schemas. Relational key columns. Easily searched via SQL, but highly resistant to evolution.

## Semi-Structured Data

Expressed via serialization languages (JSON, YAML, XML). Uses key/value pairs and tags. No static tables.

## Unstructured Data

Ambiguous organisation. Massive file sizes (photos, videos, logs, documents).

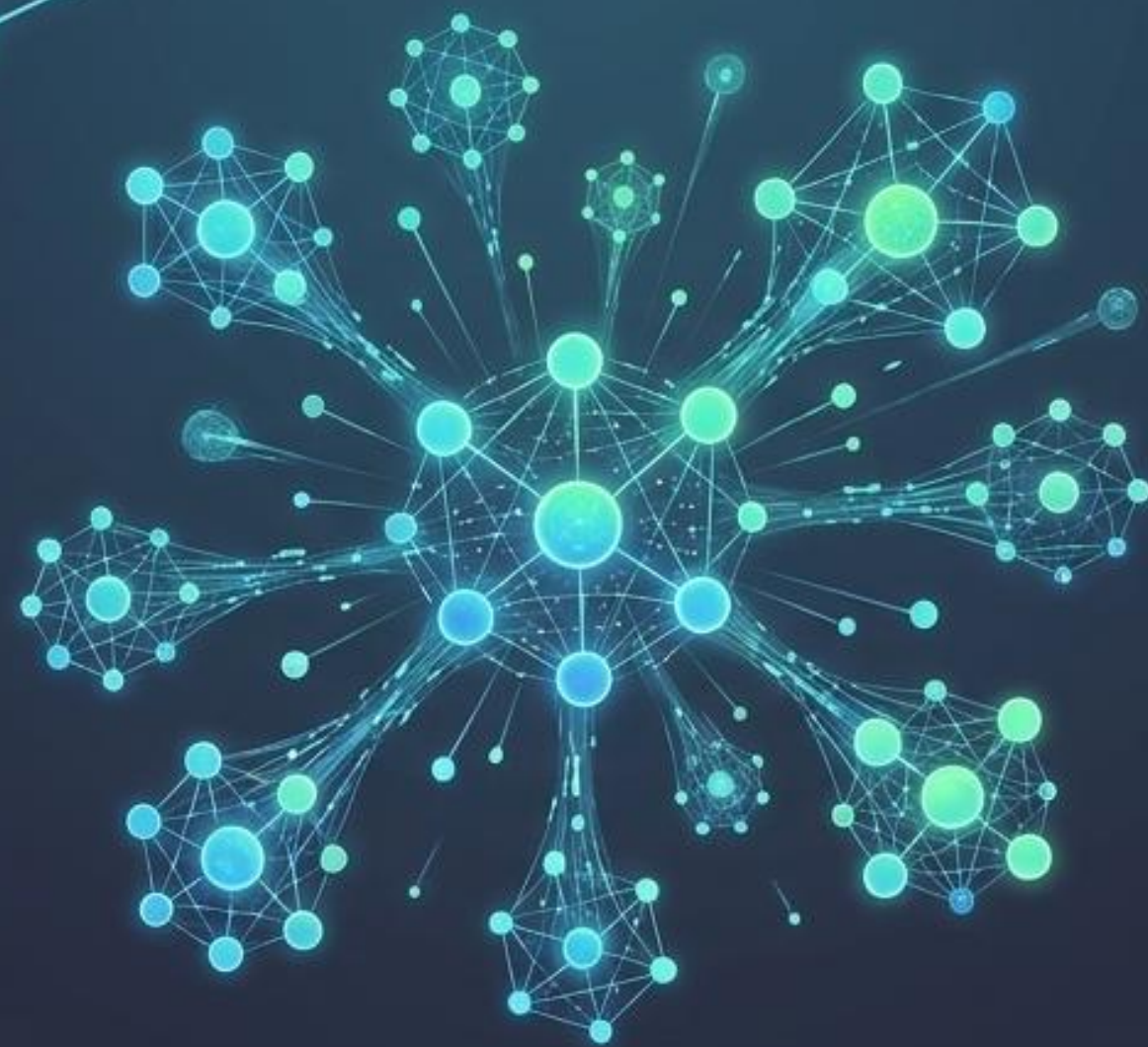


## The Relational Bottleneck

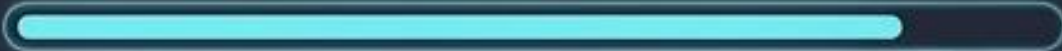
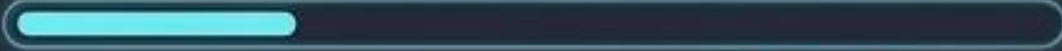
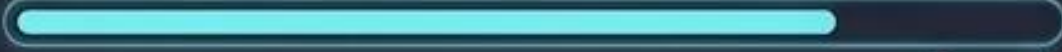
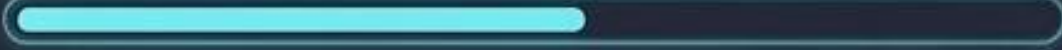
- Inflexible data structures
- Performance hindered by strict relational axioms
- Fundamental difficulty of scaling across multiple nodes

## The NoSQL Paradigm (NoREL)

- Massive scalability across multiple nodes
- Removal of explicit relational modelling
- Deliberate trade-off of strict ACIDity for eventual consistency to achieve high performance



# The Architect's Trade-off Matrix

|                     | RDBMS                                                                                                       | NoSQL                                                                                                            |
|---------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Availability        |  Good                     |  Continuous                   |
| Consistency         |  Good                     |  Eventually Consistent / Poor |
| Data Handling       |  Medium / Large workloads |  Big Data Scale               |
| Performance         |  Low at scale           |  High                       |
| Reliability         |  Good                   |  Low to Moderate            |
| Read/Write Strategy |  Scaling Reads          |  Scaling Reads and Writes   |
| Scalability         |  Moderate               |  High                       |

# The Taxonomy of NoSQL



## Key-Value Stores

- Mechanism: The Hash
- Example: Redis



## Document Databases

- Mechanism: The Nested Object
- Example: MongoDB



## Columnar Databases

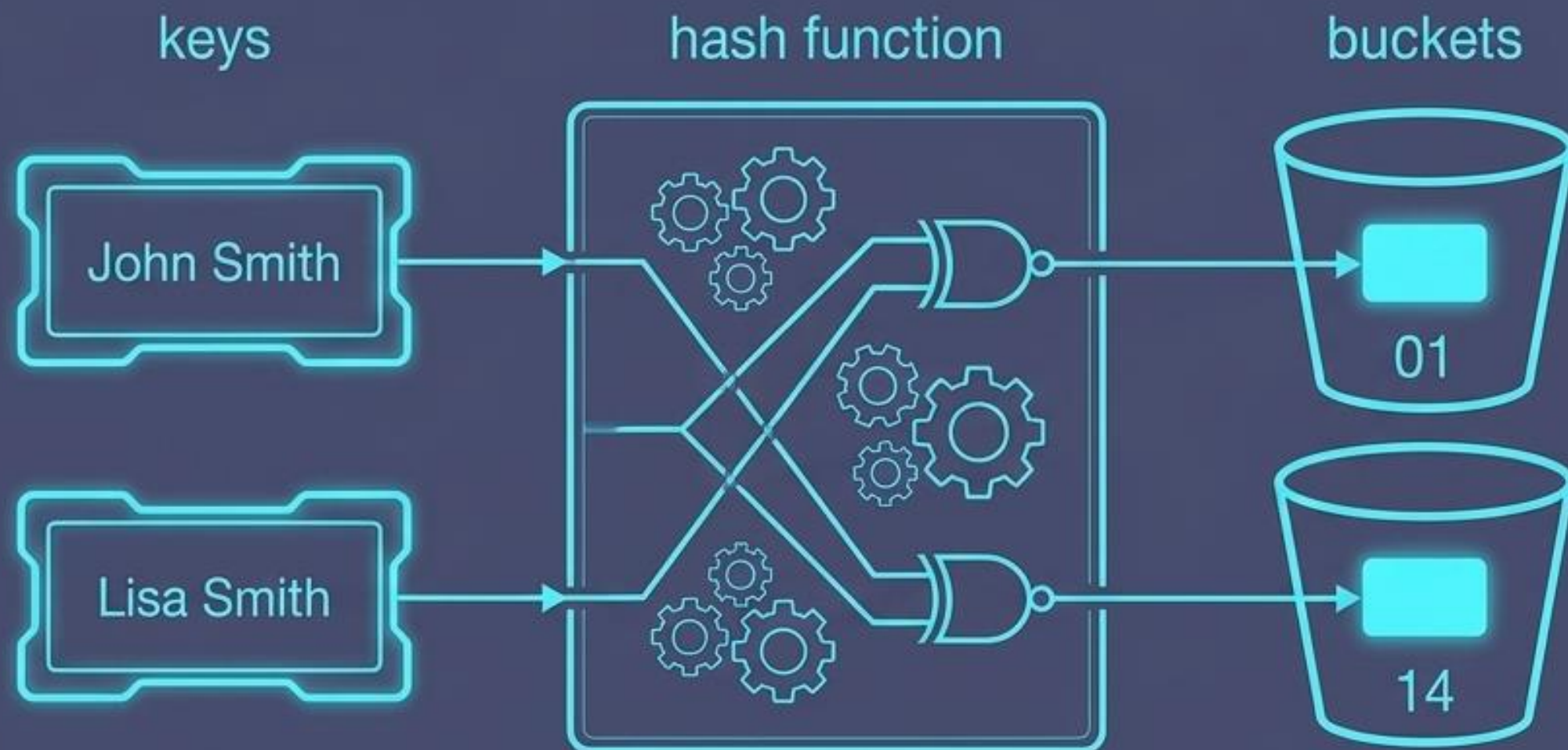
- Mechanism: The Partition
- Example: Cassandra



## Graph Databases

- Mechanism: The Relationship
- Example: Neo4j

# Key-Value Stores



## Mechanism

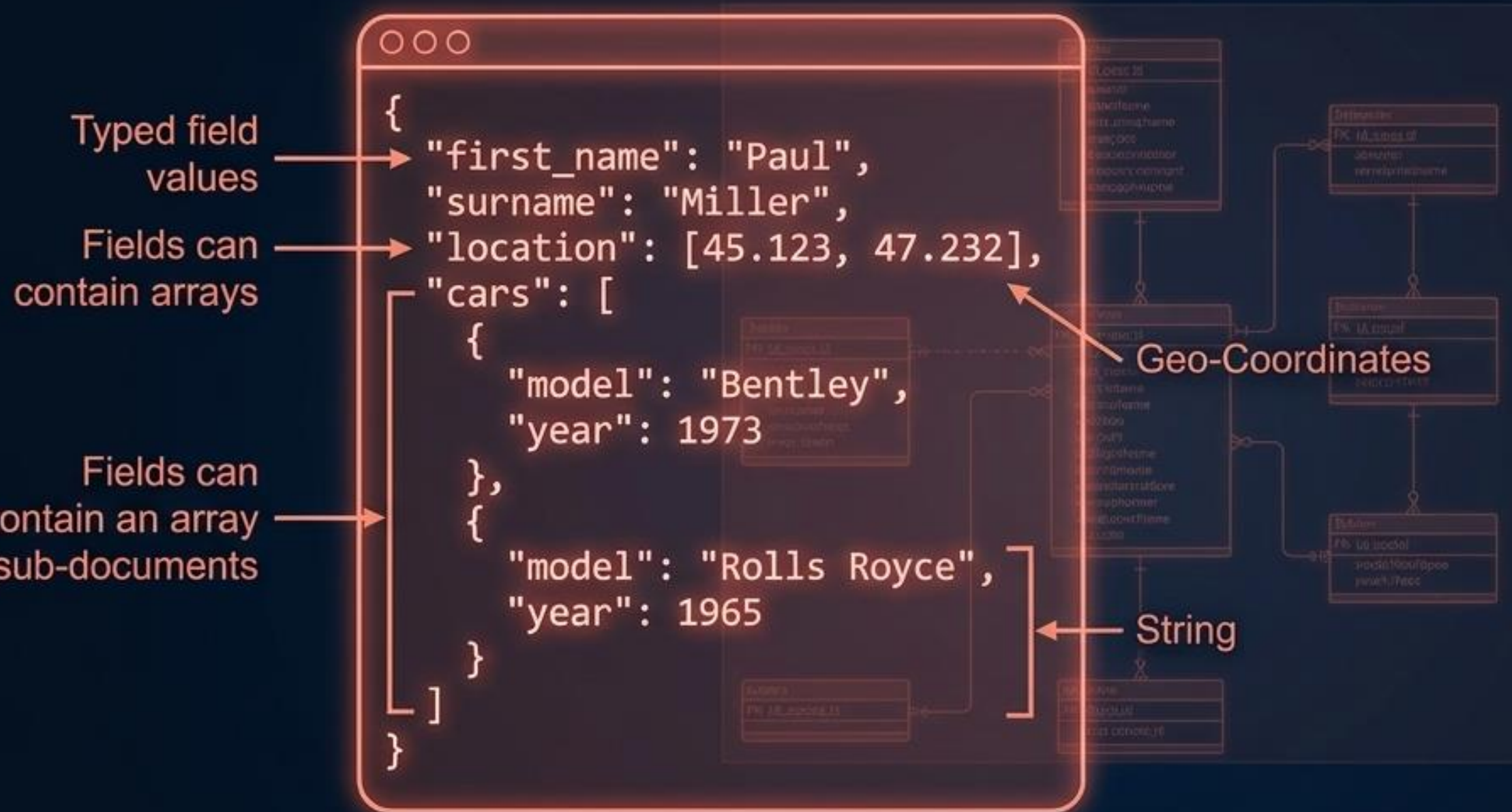
The least complex NoSQL architecture. Employs a hashing mechanism that maps indexed keys (usually strings) directly to values (opaque blobs / uninterpreted byte sequences).

## Architectural Advantage

Constant time access. Maintains blazing-fast performance and lookup speeds regardless of massive scale.

**Real-World Application:** Ideal for session info and user profiles. Powers Twitter (timeline delivery), Pinterest (follower lists), Coinbase (rate limiting), and Quora (caching).

# Document Databases



## Mechanism:

Expands on Key-Value logic but stores data as complex documents (usually JSON). Groups of documents form collections. The schema is completely flexible and defined purely by the document's contents.

## Architectural Advantage

The database engine can inspect document contents, enabling advanced retrieval processing without static table constraints.



**Real-World Application:** SEGA (11M in-game accounts), Cisco (VSRM platform scalability), Weather Channel (real-time alerts to 40M users), Aer Lingus.

# Columnar Databases



## Mechanism:

Organises data into rows and columns but without strict schema requirements (e.g., rows do not need values for all columns). Uses hashing to retrieve partitioned rows.

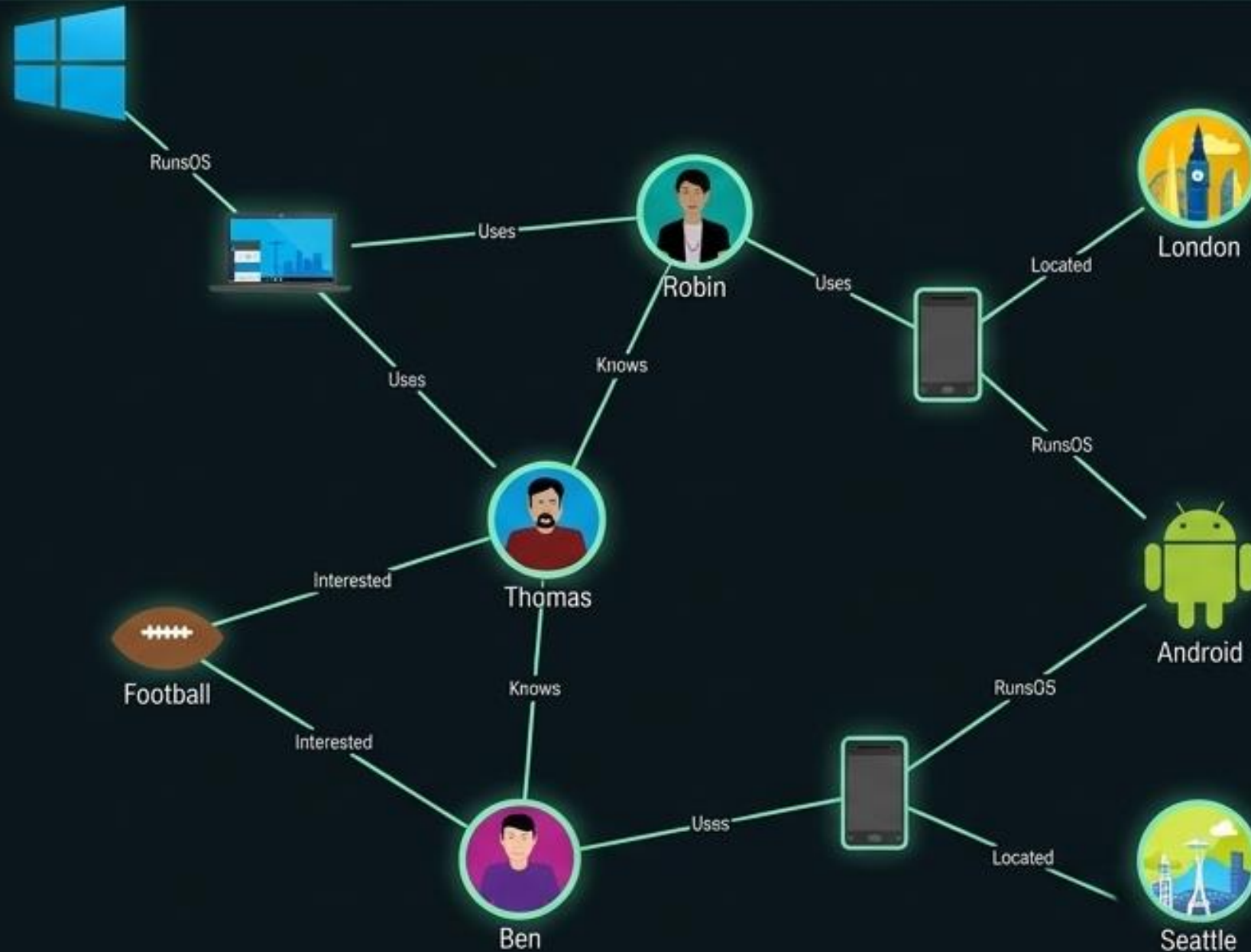
## Architectural Advantage

Extremely fast to load and query specific data attributes. Built for highly scalable architectures handling massive data volumes.



**Real-World Application:** IoT, personalization engines. Spotify (user profiles and metadata), Facebook (Messages, Nearby Friends), Outbrain (190B monthly recommendations).

# Graph Databases



## Mechanism:

Replaces the concept of tables with a literal graph metaphor. Data items are Nodes and relationships are Edges.

Nodes contain direct lists of relationship records to other nodes.

## Architectural Advantage

Eliminates the time-consuming search-match (join) operations found in relational databases. Blazing fast query performance for establishing commonalities in deeply connected, complex data sets.

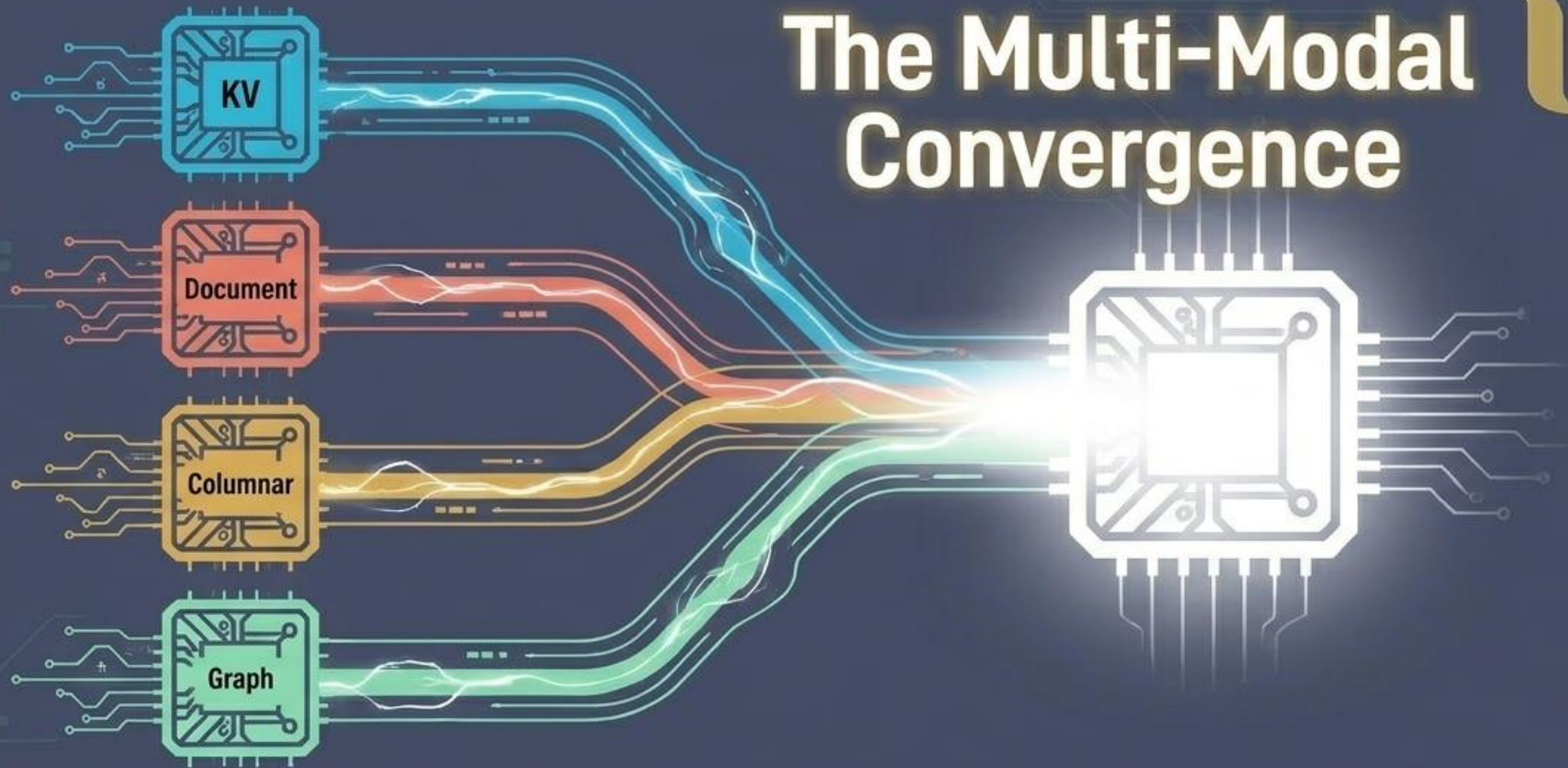


**Real-World Application:** Walmart (product recommendations), Medium (social graph personalization), Cisco (anticipating bugs), Tibco (fraud detection).

# The Architect's Selection Matrix

|                  | Key-Value                | Document                     | Columnar           | Graph                             |
|------------------|--------------------------|------------------------------|--------------------|-----------------------------------|
| Complexity Level | Lowest                   | Moderate                     | High               | Highest (Connected)               |
| Core Metaphor    | Hash Bucket              | Nested Objects               | Wide Partitions    | Nodes & Edges                     |
| Best Used For    | Quick lookups & sessions | Flexible profiles & catalogs | Heavy writes & IoT | Social networks & fraud detection |

# The Multi-Modal Convergence



**The emergence of Multi-Modal systems eliminates the need to manage fractured database architectures.**

Multi-modal databases support multiple distinct data models (KV, Document, Columnar, Graph) against a single, integrated backend.



**Architectural Advantage:** Allows enterprises to store parts of their system's data in the natively appropriate model without compounding administrative complexity or fragmenting application development.

# The Multi-Modal Engine: Azure Cosmos DB

## Engine Profile:

A fully managed, serverless NoSQL database.

## Key Metrics:

Single-digit millisecond response times.  
Elastic scale out of storage and throughput.



Table API



Cassandra



Column-family



Document



Gremlin  
Graph/Gremlin

## Management:

Takes administration entirely off developer hands with automated patching, updates, and turnkey global distribution.

## Compatibility:

Native support for open-source compatible APIs and SDKs.

# Multi-Modal Architecture at Scale

## Global Identity Scale

Citrix utilises Cosmos DB to support their Identity Platform, enabling seamless single sign-on for 400,000 organisations and 100 million global individuals.

## Real-Time Retail

ASOS deploys Cosmos DB to serve instant order updates and real-time personalized product recommendations across 15.4 million active customers.

## Complex Sensor Webs

Ulster University implements a multimodal architecture to handle the storage, processing, and analysis of billions of diverse sensor data elements.

**The modern architect does not conform data to the database; they select the database that conforms to the data.**